

UNIVERSITY OF SOUTHERN DENMARK

DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE

DM546: COMPILER CONSTRUCTION

Full Interpreter

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1 Preliminaries

If you used specific setups, not-well-known tools, or libraries briefly introduce them. Provide references to those artifacts. Note: you do not need to repeat what we discussed in the lectures and exercises.

2 Technical Description

Here you explain the technical work you have carried out. You should present the overall architecture of your interpreter. Try to explain the development choices you made taking into account important properties, such as reusability and performance.

2.1 Architecture

Please provide at least **one diagram** of the entire architecture and describe it. How do the individual elements of the interpreter (i.e., scanner, parser, etc.) work together to execute a given program? Have you applied specific design patterns?

2.2 Implementation

If you have implemented specific algorithms or used functionality not covered in the lecture, please explain it here. Which kind of design decisions did you take? Answer the following questions in the respective sections.

- How do you recognize tokens in the Scanner?
- How does the Parser work?
- How do you perform the semantic analyses?
- How do you handle the casting of types?
- How do you handle functions?
- What tasks are performed by the interpreter?
- How and when do you report errors?

2.2.1 Syntactic Analysis

2.2.2 Semantic Analysis

2.2.3 Tree-Walk Interpretation

2.2.4 Monitoring and Logging

Do you trace the state of the interpreter and its components (i.e., the scanner, parser, etc)?

2.3 Testing & Evaluation

How did you test your interpreter? Which coverage criterion did you apply?

Describe the test setup and the test inputs with their expected outputs. If you have documented this in the source code a link and submission of the generated JavaDoc is also fine.

If the provided tests have been changed or do not work, please explain here what tests fail, why and what it would require to fix them.

3 Teamwork Organization

Describe how you organized the work together in the team? How did you communicate, implement and distribute tasks?

Additionally, reflect on your individual learning outcomes (particularly if you worked alone).

4 Discussion

In this section you review possible alternative solutions to the same challenge you have tried to address in your project, or alternative methodologies that you may have followed (e.g., choice of other technologies for implementing the project). Provide a discussion on the implications of your choices in the design of your work and the technologies/techniques that you have used.

Discuss what are the shortcomings/limitations of your project, possibly explaining how they could be solved or mitigated.